

■ 2.8.4 Resource-Constrained Calculations

When you run *Mathematica* interactively, it is quite common to try a calculation, and to stop if it seems to be taking too long. If you use *Mathematica* in a batch-processing mode, you may not be able to monitor its progress, and to stop it if it is taking too long. The functions `TimeConstrained` and `MemoryConstrained`, however, allow you to try a calculation, and automatically stop it if it takes too much time or memory.

<code>TimeConstrained[expr, t]</code>	tries to evaluate <i>expr</i> , stopping after <i>t</i> seconds
<code>MemoryConstrained[expr, b]</code>	tries to evaluate <i>expr</i> , stopping if more than <i>b</i> bytes of additional memory are requested

Resource-constrained calculations.

The way that `TimeConstrained` and `MemoryConstrained` work is effectively to generate an interrupt if the constraints you specify are overrun. In many cases, *Mathematica* will not be able to stop immediately when the interrupt occurs, and so may well use up more time or memory than the constraint you gave. Whenever a calculation is interrupted, the result *Mathematica* returns is the symbol `$Interrupted`.

If you use `TimeConstrained` in programs you write, you should be sure to remember that different kinds of computers can run at very different speeds, so that `TimeConstrained` may give you different results.